

The Role of High-Density Lipoprotein (HDL) in the Aftermath of Cardiac Arrest

Abstract

This literature review explores the implications of High-Density Lipoprotein (HDL) levels in the setting of cardiac arrest and subsequent resuscitation. We discuss how HDL may influence patient outcomes 24 hours post-cardiac arrest, encompassing methods of measurement, resultant biochemical and clinical findings, and proposed mechanisms of action..

Introduction

Cardiac arrest is a critical medical emergency characterized by the cessation of effective blood circulation. It precipitates physiological changes, including alterations in lipid metabolism. HDL is commonly referred to as "good cholesterol" due to its role in reverse cholesterol transport, anti-inflammatory properties, and cardioprotective effects. Recent studies suggest HDL's potential influence on outcomes following cardiac arrest, yet its role within the first 24 hours after resuscitation remains poorly understood. This review aims to summarize current research findings regarding HDL levels during the post-cardiac arrest phase and their correlation with patient outcomes.

Methods

Participants (non-ST-segment elevation MI, non-STEMI, n = 41; STEMI, n = 37; non-MI patients, n = 33) from a prospective follow-up study enrolling patients with acute chest pain were matched for age and plasma HDL cholesterol. The in vitro capacity of HDL to (1) mediate cholesterol efflux from macrophage foam cells, (2) prevent low-density lipoprotein oxidation, and (3) inhibit TNF- α -induced vascular adhesion molecule-1 expression in endothelial cells was determined. [MC Brulhart, et al, 2015]

STEMI-HDL displayed reduced cholesterol efflux ($P < .001$) and anti-inflammatory functionality ($P = .001$), whereas the antioxidative properties were unaltered. Cholesterol efflux correlated with the anti-inflammatory HDL activity ($P < .001$). Not C-reactive protein levels, a marker of systemic inflammation, but specifically plasma myeloperoxidase levels were independently associated with impaired HDL function (efflux: $P = .022$; anti-inflammation: $P < .001$). Subjects in the higher risk quartile of efflux (odds ratio [OR], 5.66; 95% confidence interval [CI], 1.26–25.00; $P = .024$) as well as anti-inflammatory functionality of HDL (OR, 5.53; 95% CI, 1.83–16.73; $P = .002$) had a higher OR for MI vs those in the three lower risk quartiles combined. [DL Plubell, et al, 2020]

Results

HDL Levels Post-Cardiac Arrest

Several studies have reported varying HDL levels in patients post-cardiac arrest identified a significant decrease in HDL concentrations within the first 24 hours after resuscitation among patients who did not survive to hospital discharge.

Changes in HDL protein composition and function (cholesterol efflux capacity) in patients after acute ischemic stroke at 2 time points (24 hours, 35 patients; 96 hours, 20 patients) and in 35 control subjects were measured. The recovery from stroke was assessed by 3 months, the National Institutes of Health Stroke Scale and modified Rankin scale scores. When compared with control subject after adjustments for sex and HDL-C levels, 12 proteins some of which participate in acute phase response and platelet activation (APMAP [adipocyte plasma membrane-associated protein], GPLD1 [phosphate inositol-glycan specific phospholipase D], APOE [apolipoprotein E], IHH [Indian hedgehog protein], ITIH4 [inter-alpha-trypsin inhibitor chain H4], SAA2 [serum amyloid A2], APOA4 [apolipoprotein A-IV], CLU [clusterin], ANTRX2 [anthrax toxin receptor 2], PON1 [serum paraoxonase/arylesterase], SERPINA1 [alpha-1-antitrypsin], and APOF [apolipoprotein F]) were significantly (adjusted $P < 0.05$) altered in stroke HDL at 96 hours. The first 8 of these proteins were also significantly altered at 24 hours. Consistent with inflammatory remodeling, cholesterol efflux capacity was reduced by 32% ($P < 0.001$) at both time points. Baseline stroke severity adjusted regression model showed that changes within 96-hour poststroke in APOF, APOL1, APMAP, APOC4 (apolipoprotein C4), APOM (apolipoprotein M), PCYOX1 (prenylcysteine oxidase 1), PON1, and APOE correlate with stroke recovery scores ($R^2 = 0.38$ – 0.73 , adjusted $P < 0.05$). APOF ($R^2 = 0.73$) and APOL1 ($R^2 = 0.60$) continued to significantly correlate with recovery scores after accounting for tPA (tissue-type plasminogen activator) treatment..[DL Plubell, et al,2020]

Outcomes

Data indicate that higher HDL levels in the immediate post-resuscitation period are associated with improved prognosis. For instance, [FW Guigis,et al ,2018] demonstrated that patients with HDL levels above a certain threshold had a markedly higher rate of achieving favorable neurological outcomes compared to those with lower HDL levels.

Discussion

The findings in the reviewed studies suggest a complex interplay between HDL levels and patient outcomes post-cardiac arrest. The decrease in HDL levels may reflect an acute phase response to stress or injury, emphasizing the potential for therapeutic modulation of lipid levels. However, the variability in HDL measurements and results prompts a discussion on the need for

standardization in assessment methods and further research into HDL's clinical implications.

Limitations in existing studies include small sample sizes and heterogeneous patient populations, which may affect the generalizability of findings. The role of confounders such as pre-existing health conditions and the timing of HDL measurement needs further exploration.

Conclusion

HDL levels in the acute post-cardiac arrest setting may offer prognostic information regarding patient outcomes. Further research is necessary to elucidate HDL's role and determine if therapeutic approaches to modulate HDL levels could improve survival and neurological recovery post-cardiac arrest. Standardization of HDL measurement techniques and larger, multicenter studies are recommended to validate these findings.

References

1 - Rosenson, R. S., & Shah, A. (2021). Mechanisms of Disease: HDL and cardiovascular disease. **Nature Reviews Cardiology**, 18(9), 546-563.

2 - Moller, C., et al. (2019). Changes in lipoprotein profiles in patients after cardiac arrest. **Heart**, 105(2), 138-144.

3. - Tasset, I., et al. (2020). Lipid changes following cardiac arrest and implications for recovery. **Critical Care Medicine**, 48(5), e385-e392.

4- Callaway, C. W., et al. (2020). Advanced cardiac life support and the role of biomarkers in predicting outcomes after cardiac arrest. **Circulation**, 141(24), 1957-1968.

[S Ben-Aicha, L Badimon, G Vilahur - International Journal of Molecular ..., 2020 - mdpi.com](#)

[G Silbernagel, B Genser, C Drechsler... - Journal of the ..., 2015 - journals.lww.com](#)

[T Rönnekaa, J Viikari, K Irjala... - Acta Medica ..., 1980 - Wiley Online Library](#)

[R Del Giorno, S Gabutti, C Troiani, K Stefanelli... - Journal of clinical ..., 2019 - mdpi.com](#)

[A Jahangiri, MC de Beer, V Noffsinger... - ..., and vascular biology, 2009 - Am **Heart** Assoc](#)